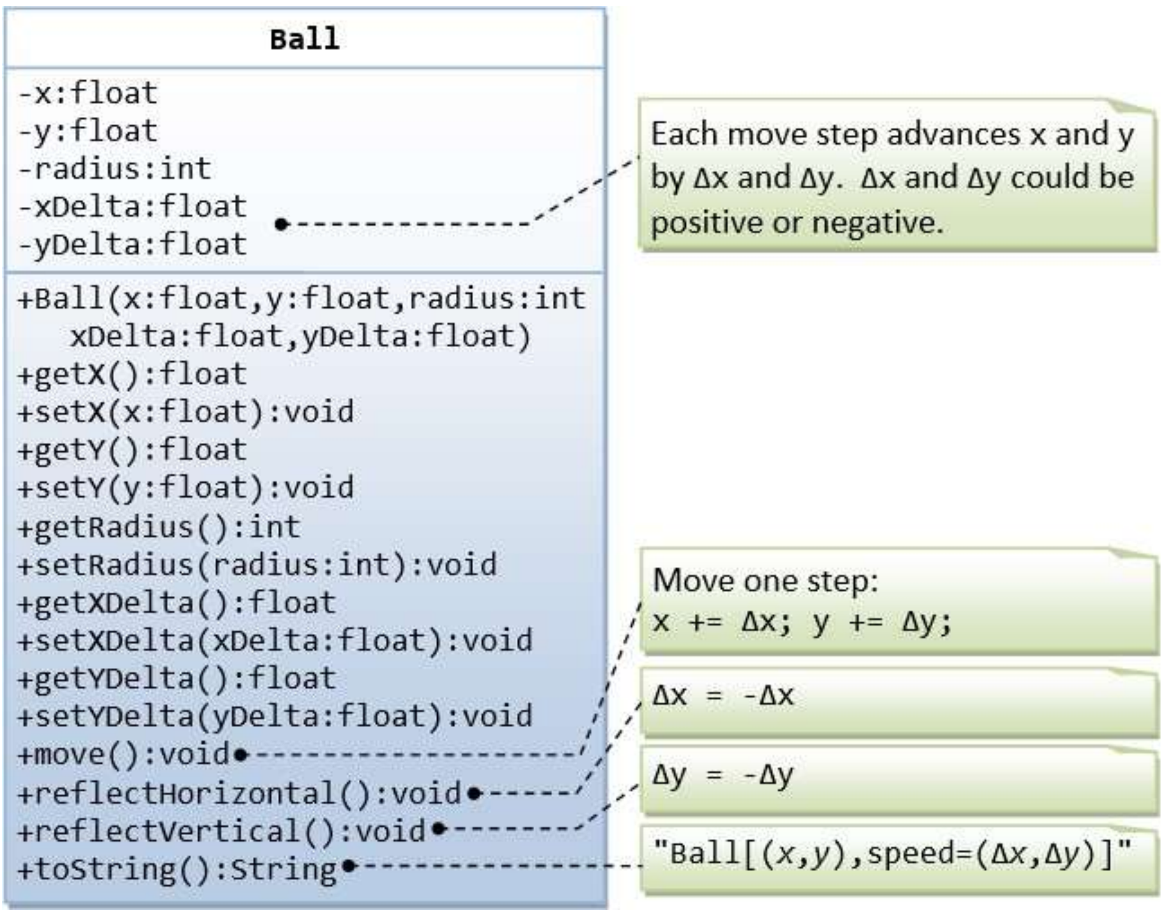
OOP Workshop

Task 1:

A class called `Ball`, which models a bouncing ball, is designed as shown in the following class diagram. It contains its radius, x and y position. Each move-step advances the x and y by delta-x and delta-y, respectively. delta-x and delta-y could be positive or negative. The `reflectHorizontal()` and `reflectVertical()` methods could be used to bounce the ball off the walls. Write the `Ball` class. Study the test driver on how the ball bounces.



Task 2:

Create an employee class that has:

- His data: id, name, email, age, salary
- default constructor and parameterized constructor that initialize all data.
- Setter
- Getter

- Display info as a virtual method
- Pure virtual method calculateSalary()

Create Manager class that inherit from Employee class has additional attributes bonus.

Create Engineer class that inherit from Employee class has additional attributes projectCount.
NOTE: (Assuming each project contributes $1000 to the salary)

Override a CalculateSalary() method in both Manager and Engineer classes to calculate their salaries, including bonuses for managers and additional project-based pay for engineers.

Task 3:

Create Duration class allows you to create instances by specifying hours, minutes, and seconds.

- A constructor to initialize the hours, minutes, and seconds.
- A Constructor that takes duration in milliseconds.
- Using Copy Constructor.
- Overloaded + operator to add two Duration.
- Overloaded - operator to subtract Duration.
- Overloaded > operator to compare two Duration.
- Overloaded < operator to compare two Duration.
- Overloaded == operator to compare two Duration.

Task 4:

Create a template class “Math” that provide several static methods:

- Pow method.
- Abs method.
- Sqrt method: note if the user enters negative number exception is thrown so, handle this scenario.